Factor Dispersion and Active-Risk Budgeting: Evidence from Factor Momentum

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Abstract

A factor-dispersion signal is useful to allocators when it improves risk awareness beyond conventional forecasts. We define a Factor Dispersion Index (FDI) from seven monthly U.S. factor returns divided by their prior 36-month volatility. High FDI is followed by a roughly 50% wider standardized factor-return range than low FDI, and its information survives controls for short-horizon volatility. In the same 657 months, adding FDI to trailing tracking error improves the fit for next-month absolute active returns from 15.1% to 17.6%. FDI does not reliably identify higher expected momentum returns: the momentum gate trails an unconditional momentum tilt, while both a constant tracking-error budget and its FDI-adjusted counterpart earn about 12.06% net annually. Their returns are virtually identical. The momentum advantage over risk balance is not statistically established in Sharpe terms and largely weakens without betting against beta. FDI is best used to reassess potential active-risk variability, not to switch factor timing on or off.

Keywords: factor investing; return dispersion; factor momentum; active allocation; risk budgeting; portfolio construction

Highlights

- Volatility-scaled factor dispersion predicts the width of next-month factor outcomes beyond short-horizon realized volatility.
- FDI relates more strongly to active-risk amplitude than to a reliably higher expected factor-momentum premium.

- A constant tracking-error budget accounts for nearly all of the additional return from a state-adjusted TE strategy; BAB concentration also matters.

Introduction

Active factor allocation has an awkward information problem. A portfolio manager can observe which factor has recently won, which one looks inexpensive, and which sleeve dominates current risk, but none of those observations answers a more basic question: how widely might the outcomes of choosing among factors differ over the next month? When outcomes across factor sleeves are tightly clustered, even a correct ranking may not pay enough to cover turnover, model error, and governance cost. When outcomes spread out, the same ranking skill has more room to create value. The missing state variable is not expected return by itself. It is the width of the opportunity set.

Equity managers have long used cross-sectional stock-return dispersion to judge whether security selection has room to work. Solnik and Roulet (2000) connect dispersion to the common movement of markets, while Stivers and Sun (2010) show that stock-level dispersion is associated with time variation in value and momentum premiums. Shalen (2025) brings the idea directly into portfolio practice in *The Journal of Portfolio Management*, arguing that the roots of stock dispersion matter for understanding active opportunity. The extension to factor sleeves is natural but not mechanical. A factor portfolio is already a long-short return source, and its volatility scale can differ sharply from that of another factor. Market, momentum, and betting against beta can have much larger monthly swings than profitability or investment. The standard deviation of seven raw factor returns will therefore be dominated by the high-volatility sleeves even if they contain no superior selection opportunity.

We address that comparability problem with a Factor Dispersion Index (FDI). Each month, every factor return is divided by a volatility estimate available before that return is observed. FDI is the cross-sectional standard deviation of the resulting standardized returns. It is high when the factor menu delivers widely separated risk-adjusted outcomes and low when those outcomes cluster. The index contains no sign about what to buy. It measures the possible amplitude of a ranking decision, not the expected alpha or the payoff per unit of active risk.

That distinction connects dispersion to the factor-momentum literature. Gupta and Kelly (2019) document momentum across a broad set of factors, and Ehsani and Linnainmaa (2022) show that persistence in factor returns is central to conventional momentum. These studies provide a directional rule: recent factor winners tend to continue outperforming recent losers. FDI supplies a different piece of information. If factor momentum identifies direction, FDI may indicate the likely breadth between winners and losers and therefore the active-risk budget worth assigning to the signal. Reversal provides a clean competing hypothesis because dispersion could instead reflect overshooting that subsequently corrects.

Using U.S. market, size, value, profitability, investment, momentum, and betting-against-beta returns from July 1963 through March 2026, we document persistent factor dispersion. After high-FDI observations, the next-month standardized factor-return range averages 3.359 versus 2.206 after low readings—approximately 52% wider. The relationship persists after controls for average lagged factor volatility, a short-horizon volatility surprise and, in a shorter sample, VIX. In a decision-relevant test, a simple trailing tracking-error measure alone explains 15.1% of the next month's absolute active return; adding FDI raises the in-sample explanatory power to 17.6% on identical observations. This incremental fit does not establish an out-of-sample risk forecast.

Factor momentum supplies an independent directional candidate, but a larger FDI reading does not make it consistently more profitable. The rank-weighted long–short momentum diagnostic earns positive average spreads after low, middle and high FDI; the middle-state estimate is imprecise and the high-minus-low difference cannot be distinguished from zero. FDI instead predicts the absolute size of next-month momentum payoffs. For a long-only momentum tilt, net active return is 2.90%, 1.91%, and 3.41% annually in low, middle, and high states, without a statistically established high-versus-low return advantage. Annualized tracking error rises more steadily, from 3.31% to 4.27% and 7.13%.

The portfolio comparison points to risk budgeting rather than a return-timing rule. From July 1971 to March 2026, risk balance earns 8.72% net annually (Sharpe 1.17), unconditional momentum earns 11.47% (1.31), and an FDI gate earns 10.04% (1.22). Gating sacrifices exposure to momentum in low-FDI months; turnover differences explain only a few basis points of the return gap. A constant 4% tracking-error target earns 12.06% (Sharpe 1.30), and adjusting that forecast using FDI also earns 12.06% (Sharpe 1.32). Thus the apparent return improvement over momentum comes from TE scaling, not FDI. Bootstrap comparisons do not establish a Sharpe advantage for unconditional momentum over risk balance and indicate that the gate is worse than unconditional momentum. Removing BAB reduces the momentum Sharpe to 1.10, close to the six-factor risk-balanced Sharpe of 1.07.

This result sits comfortably with the practical caution in factor-timing research. Asness et al. (2017) show why contrarian timing can look persuasive in theory and disappoint in implementation. DeMiguel, Garlappi, and Uppal (2009) demonstrate how estimation error can defeat optimized allocation rules, while Cederburg et al. (2020) document the instability of apparently successful dynamic risk management across assets and samples. Our rules are deliberately fixed, transparent, and sparse. The paper asks whether dispersion contains decision-relevant information, not whether a heavily optimized FDI map can be made to dominate in sample.

The contribution is a decision framework with three distinct inputs. FDI indicates how widely relative outcomes may differ and provides an additional check against trailing tracking error. A separate return signal proposes portfolio direction, while the investor's liquidity, financing and tracking-error capacity determine the permitted active exposure. The evidence favors monitoring and testing these components separately rather than treating FDI as a mechanical authorization to increase or suppress risk.

Data and the Factor Opportunity Set

The opportunity set contains seven recognizable U.S. factor sleeves. Monthly excess-market, size, value, profitability, and investment returns come from the Kenneth R. French Data Library. Momentum comes from its companion monthly momentum file. Betting-against-beta returns come from the AQR Data Library and follow the financing-constraint mechanism developed by Frazzini and Pedersen (2014). Daily VIX observations come from the Federal Reserve Bank of St. Louis FRED database and are averaged within calendar months. The common factor sample contains 753 monthly observations from July 1963 through March 2026. VIX is available from January 1990.

These seven factors span several of the most established sources of systematic return. MKT is the excess return on the broad equity market. SMB and HML capture size and value in the Fama and French (1993) framework. RMW and CMA represent profitability and investment in the five-factor model of Fama and French (2015). MOM is the continuation effect associated with Carhart (1997), and BAB is a defensive premium linked to leverage constraints. The set is not intended to represent the entire factor zoo. Harvey, Liu, and Zhu (2016) explain why a large search over candidate factors requires a high statistical hurdle, and McLean and Pontiff (2016) show that published anomalies often weaken after discovery. A short list of established sleeves limits exposure to factor discovery and backtest-selection concerns raised by Arnott et al. (2019). Any live FDI must be recomputed over the investor's own feasible sleeves; published research portfolios are not necessarily replicable net of their internal trading costs.

Exhibit 1 reports the unconditional characteristics. Annualized volatility ranges from 7.16% for CMA and 7.70% for RMW to 15.44% for MKT and 14.44% for MOM. That scale difference is central to the FDI design. A raw cross-sectional standard deviation would allow MKT and MOM to dominate the reading simply because a one-standard-deviation month is larger in percentage-return units. The return distributions are also asymmetric. MOM has skewness of -1.31 and a worst month of -34.34%, consistent with the momentum-crash evidence of Daniel and Moskowitz (2016). Every sleeve suffers a meaningful peak-to-trough drawdown; the maximum exceeds 54% for MKT, SMB, HML, MOM, and BAB. Factor labels do not remove tail risk.

Exhibit 1. Factor Data and Unconditional Characteristics

Factor	Annual Return	Annual Volatility	Sharpe Ratio	Skewness	Worst Month	Maximum Drawdown	Start	End	Months
MKT	7.02%	15.44%	0.45	-0.50	-23.19%	-55.75%	1963-07-31	2026-03-31	753
SMB	2.22%	10.47%	0.21	0.36	-15.54%	-56.47%	1963-07-31	2026-03-31	753
HML	3.58%	10.29%	0.35	0.09	-13.83%	-57.79%	1963-07-31	2026-03-31	753
RMW	3.16%	7.70%	0.41	-0.33	-18.95%	-41.78%	1963-07-31	2026-03-31	753
CMA	2.98%	7.16%	0.42	0.27	-7.08%	-27.31%	1963-07-31	2026-03-31	753
MOM	7.17%	14.44%	0.50	-1.31	-34.34%	-57.82%	1963-07-31	2026-03-31	753
BAB	9.05%	11.19%	0.81	-0.42	-15.68%	-54.65%	1963-07-31	2026-03-31	753

Notes: Monthly U.S. factor returns, July 1963-March 2026; annualized return and risk measures unless noted. Maximum drawdown is calculated from compounded monthly factor returns.

The heterogeneity has a direct allocation implication. A portfolio manager who compares raw factor returns is implicitly assigning more "importance" to volatile sleeves. A ranking based on raw 12-month return can make the same error. Barroso and Santa-Clara (2015) and Moreira and Muir (2017) show why volatility scaling can materially alter the behavior of dynamic strategies. Our use is narrower: standardization makes contemporaneous outcomes comparable across factors before dispersion is measured. It does not presume that volatility is fully predictable or that scaling automatically raises expected utility.

Constructing the Factor Dispersion Index

Let $r_{i,t}$ denote the return on factor i in month t. Let $\hat{\sigma}_{i,t-1}$ be its monthly volatility estimated from the 36 returns ending in month $t-1$. The standardized factor return is

$$z_{i,t} = \frac{r_{i,t}}{\hat{\sigma}_{i,t-1}}.$$

Because the denominator uses information ending before month t, an unusually large standardized outcome cannot reduce its own measured magnitude by inflating the volatility estimate. We require the full 36-month window. The baseline balances responsiveness with enough observations to avoid unstable monthly scales; 24- and 60-month windows are reported as robustness checks.

For $N = 7$ factors, FDI is

$$\mathrm{FDI}_t = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}\left(z_{i,t} - \bar{z}_t\right)^2}, \qquad \bar{z}_t = \frac{1}{N}\sum_{i=1}^{N} z_{i,t}.$$

Mean-centering is important. A month in which all factors earn the same positive standardized return can be rewarding for a long factor portfolio but offers little cross-factor selection opportunity. Conversely, a month in which some factors gain and others lose can produce a high FDI even if the average factor return is near zero. FDI is therefore not factor-portfolio performance, factor volatility, or average absolute return. It is the width of standardized outcomes around their cross-sectional mean.

Exhibit 2 summarizes the measure. From July 1966 through March 2026, FDI averages 0.997, has a median of 0.875, and ranges from 0.160 to 4.535. Its 5th and 95th percentiles are 0.383 and 2.026. The standardized-return correlation matrix in Panel B makes clear that factor dependence remains present after volatility scaling. HML and CMA correlate 0.67, while MKT correlates -0.36 with CMA and -0.21 with HML. FDI does not attempt to remove correlation; it records the cross-sectional outcome generated by the full covariance structure in a particular month.

Exhibit 2. Constructing the Factor Dispersion Index

Panel A: Distribution of FDI

Statistic	FDI
Mean	0.997
Median	0.875
Std. Dev.	0.540
5th Percentile	0.383
95th Percentile	2.026

Minimum	0.160
Maximum	4.535
Lag-1 Autocorrelation	0.333

Panel B: Correlations of volatility-standardized factor returns

index	MKT	SMB	HML	RMW	CMA	MOM	BAB
MKT	1.000	0.288	-0.210	-0.105	-0.359	-0.095	0.092
SMB	0.288	1.000	0.000	-0.294	-0.101	-0.116	0.056
HML	-0.210	0.000	1.000	-0.116	0.666	-0.296	0.208
RMW	-0.105	-0.294	-0.116	1.000	-0.154	0.076	0.007
CMA	-0.359	-0.101	0.666	-0.154	1.000	-0.107	0.210
MOM	-0.095	-0.116	-0.296	0.076	-0.107	1.000	0.140
BAB	0.092	0.056	0.208	0.007	0.210	0.140	1.000

Notes: FDI is the cross-sectional standard deviation of factor returns divided by their lagged 36-month volatility.

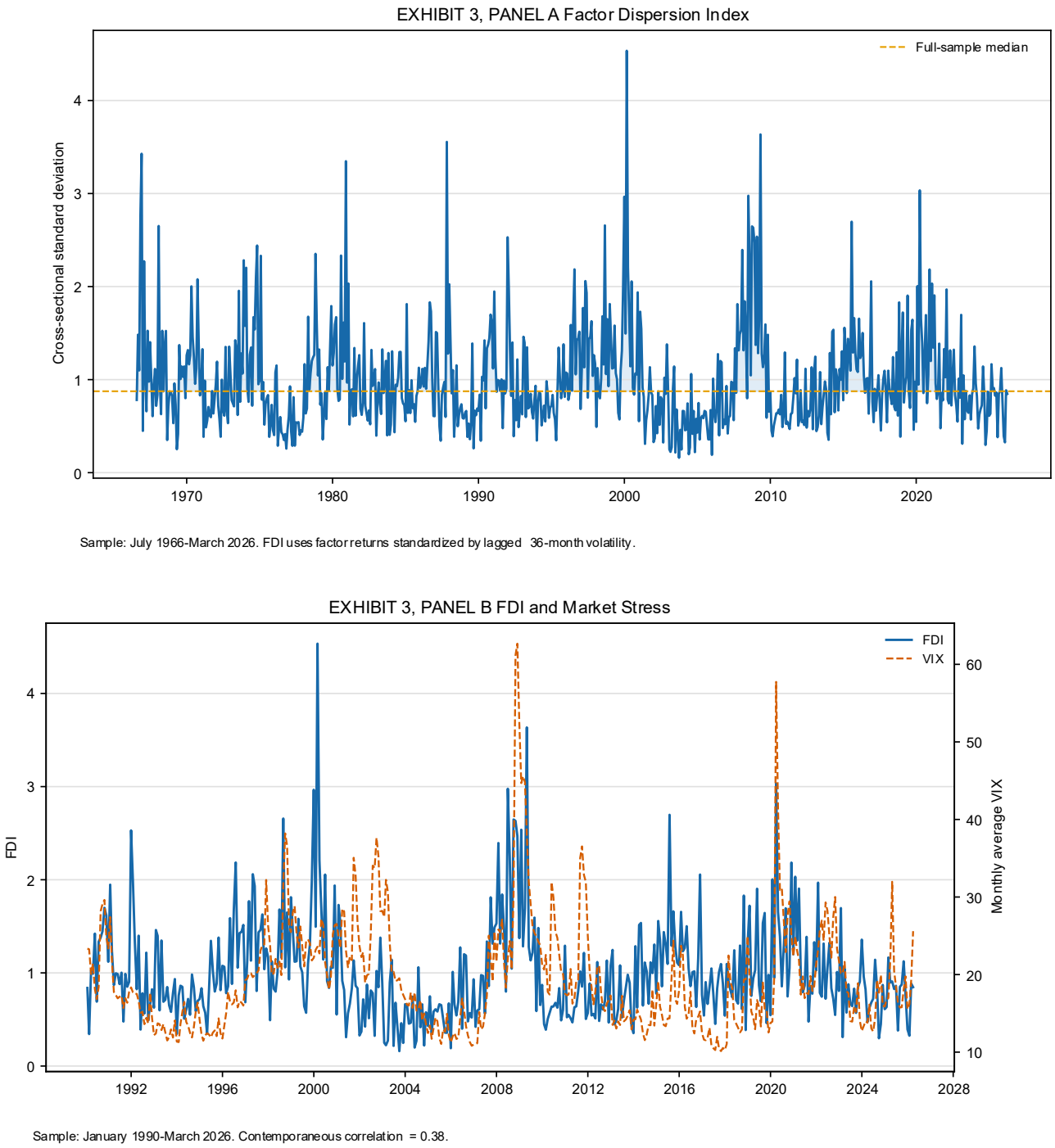
To define investable states without hindsight, we calculate the percentile of each FDI observation relative to the expanding history available at that date. State classification begins after 60 FDI observations. Readings below the 33rd percentile are low, readings from the 33rd to the 67th percentile are middle, and readings at or above the 67th percentile are high. No observation is classified using a threshold estimated from future data. At each month-end, lagged factor volatility, the current expanding percentile and the 12-month factor-momentum ranking are all observable; weights determined at that date are held against subsequent-month returns. All strategy thresholds and tilts are fixed across evaluation periods, and return tables exclude terminal states lacking a subsequent month.

Exhibit 3, Panel A plots the full history. FDI is episodic rather than smoothly trending. Large spikes appear around the mid-1970s, the early 1980s, the 1987 market break, the technology unwind, the global financial crisis, and the 2020 pandemic shock, but high readings are not confined to recession dates. The index is a realized cross-factor outcome measure, so it can rise because factors respond differently to a common shock or because sleeve-specific returns separate.

Panel B compares FDI with the monthly average VIX after 1990. Their contemporaneous correlation is 0.38. Market stress clearly matters, but the overlap is incomplete. VIX prices near-term equity-index uncertainty; FDI measures the realized cross-sectional distance among standardized factor returns. A common market selloff can raise VIX while leaving factor-relative outcomes close together. A rotation between value, momentum, defensive, and investment styles

can raise FDI without producing an extreme VIX reading. A portfolio dashboard has room for both.

Exhibit 3. Factor Dispersion through Time



Notes: Panel A shows FDI from July 1966-March 2026. Panel B compares FDI with monthly average VIX from January 1990.

Dispersion Is Opportunity, Not Direction

The interpretation of FDI requires discipline. Dispersion can make a correct active choice more valuable, but it cannot make the choice correct. If one factor earns two standardized units and another loses two, the cross-sectional opportunity is large. FDI does not reveal whether the winner will continue, reverse, or become irrelevant next month. Treating high dispersion as an instruction to buy recent losers would silently impose a reversal model. Treating it as an instruction to buy recent winners would silently impose momentum. The index itself supports neither direction and does not establish the expected return per unit of tracking error.

We use factor momentum because it is economically familiar, easy to implement, and supported by existing evidence. Jegadeesh and Titman (1993) establish continuation in stock returns, Moskowitz, Ooi, and Pedersen (2012) document time-series momentum across markets, and Asness, Moskowitz, and Pedersen (2013) show that value and momentum recur across asset classes. Gupta and Kelly (2019) and Ehsani and Linnainmaa (2022) move the logic to factor portfolios. The alternative reversal rule is useful precisely because it has the same information set and opposite direction.

At the end of month t, the score for factor i is its compounded return over the latest 12 months divided by its current annualized 36-month volatility:

$$m_{i,t} = \frac{\prod_{k=0}^{11}\left(1 + r_{i,t-k}\right) - 1}{\sqrt{12}\,\hat{\sigma}_{i,t}}.$$

Factors are ranked from one to seven on this score. The centered rank $q_{i,t}$ ranges from -1 for the lowest score to +1 for the highest. Returns in month $t + 1$ are used to evaluate the ranking. There is no overlap between the information used to rank and the return used to judge the decision.

The momentum-selection diagnostic assigns positive weights to higher-ranked factors and negative weights to lower-ranked factors using all seven centered ranks. These ranks are multiplied by inverse trailing volatility, and the total absolute weight is normalized to two. The resulting long–short series is not a top-two-minus-bottom-two-only strategy; the top-versus-bottom-two factor-level sorts in Exhibit 7 are separate descriptive tests. Reversal is the exact negative of the same diagnostic, rather than an independently estimated strategy. Long-only allocations translate the centered ranks into nonnegative sleeve tilts around the strategic risk-balanced portfolio.

Does FDI Forecast the Width of Future Outcomes?

Exhibit 4 provides the most direct state test. Low-FDI observations have an average current FDI of 0.531. Their average next-month FDI rises to 0.790, partly reflecting mean reversion from a compressed starting point. Middle observations start at 0.892 and are followed by 0.974. High observations start at 1.577 and are followed by 1.213. The high state mean-reverts, yet its next-month dispersion remains 54% greater than the low state's. The same ordering appears in the standardized range, which is 3.359 after high FDI, 2.690 after middle FDI, and 2.206 after low FDI.

Exhibit 4. What Follows Different FDI States?

Next-month cross-factor opportunity and directional selection payoffs, classified using expanding real-time FDI terciles.

FDI State	Months	Current FDI	Next-Month FDI	Next-Month Standardized Range	Momentum Selection Spread (Annualized)	Momentum Spread t-Statistic	Reversal Selection Spread (Annualized)	Risk-Balanced Return (Annualized)
Low	234	0.531	0.790	2.206	8.36%	4.12	-8.36%	8.73%
Middle	210	0.892	0.974	2.690	4.24%	1.56	-4.24%	8.92%
High	213	1.577	1.213	3.359	9.24%	2.30	-9.24%	8.60%

Panel B: Factor-momentum spread and FDI percentile (Newey–West, three lags).

Specification	Months	Slope (monthly)	HAC t	R^2
Signed spread on FDI percentile	657	0.0000	0.00	0.0%
Absolute spread on FDI percentile	657	0.0199	3.32	4.4%
High-minus-low momentum spread (monthly)	447	0.0007	0.20	0.0%

Notes: Panel A — FDI states are dated at the decision month-end and include only the 657 decisions with observed subsequent-month outcomes (234 low, 210 middle, 213 high). The selection spread uses all seven centered risk-adjusted ranks, scaled by inverse trailing volatility; reversal is its arithmetic negative. Panel B — The signed and absolute outcomes are next-month rank-weighted long–short selection returns. A high-minus-low indicator uses only those two terciles; the slope is a monthly return difference. State classification uses current information and next-month observations are never included in the predictor.

Higher FDI tends to be followed by a wider standardized cross-factor range even after the current dispersion state partly mean-reverts. More relevant for risk budgeting, it adds information about the absolute size of the next month's active return to an observable trailing tracking-error measure. These are predictive associations, not expected-alpha estimates; Exhibit 5 separates standardized and raw outcomes to make comparisons with unscaled dispersion fair.

Exhibit 5 compares current FDI with next-month dispersion and standardized factor-return ranges. The forecast relationship remains after controlling for average lagged factor volatility and, from 1990, VIX. Panel B asks whether this finding simply reflects persistent short-run volatility relative to a slow 36-month scale. Panel C asks the more operational question: does FDI add explanatory information beyond the tracking error a manager already monitors? Details of the three-lag heteroskedasticity-and-autocorrelation-consistent tests are reported in the exhibit and the reproducibility code.

Exhibit 5. Predicting the Width of Future Factor Outcomes

Specification	Observations	R-squared	FDI z Coefficient	FDI z t-Statistic	Average Vol z Coefficient	Average Vol z t-Statistic	VIX Average Coefficient	VIX Average t-Statistic

Next-month FDI	716	11.1%	0.180	8.496				
Next-month standardized range	716	11.0%	0.486	8.043				
Next-month range, controlling for factor volatility	716	14.3%	0.424	6.503	-0.273	-5.171		
Next-month range, VIX sample	434	25.8%	0.241	2.243	-0.425	-6.379	0.074	5.860

Panel B: Short-horizon volatility controls and matched raw-outcome benchmarks

Model	N	FDI coeff.	FDI HAC t	R^2	Control t-statistic(s)
FDI + 6m volatility	716	0.480	7.65	11.1%	Short Realized Vol t=0.38
FDI + volatility surprise	716	0.247	3.16	15.5%	Short/Slow Vol Ratio t=5.21
FDI + raw dispersion	716	0.688	7.49	12.0%	Raw Dispersion t=-3.14
Raw dispersion only	716	—	—	4.4%	Raw Dispersion t=4.80
FDI + raw dispersion + short vol	716	0.795	7.97	13.0%	Short Realized Vol t=2.85, Raw Dispersion t=-4.50
FDI on next raw range	716	0.013	5.05	7.05%	—
Raw dispersion on next raw range	716	—	—	15.92%	Raw Dispersion t=9.56

Panel C: Predicting absolute next-month active returns against trailing tracking error

	Model	Months	R^2	FDI percentile t	Trailing TE t
	FDI only	657	5.75%	3.65	—
	Trailing TE only	657	15.05%	—	3.55
TE	FDI and trailing	657	17.64%	3.75	3.52

Notes: The dependent variable in Panel A is next-month FDI or the next-month maximum-minus-minimum standardized factor outcome. Predictor z-scores make coefficient scales comparable, while implementable terciles use expanding information only. Newey–West standard errors use three monthly lags; regressions are in-sample and the VIX sample begins in 1990. Panel B's first five specifications predict next-month standardized factor-return range; the last two predict the next-month range of unscaled monthly factor returns. The raw-outcome comparison places FDI and raw dispersion on the same 716 dates. The first five specifications also use FDI-eligible observations, and estimates are not directly comparable across different dependent variables. Six-month realized volatility and the six-month/lagged-36-month volatility ratio are observable at month-end t. Panel C — The outcome is the absolute next-month net active return of unconditional momentum relative to risk balance. The risk-model comparator is the trailing 12-month realized tracking error. All three specifications use the same 657 months; these in-sample fits are diagnostic and not calibrated out-of-sample forecasts.

For portfolio managers, Panel C offers the most direct comparison: trailing tracking error alone accounts for 15.1% of the variation in next-month absolute active return, compared with 17.6% when FDI percentile is included, on the same 657 decisions. Panel B provides an important qualification. FDI still predicts next-month standardized dispersion when recent realized volatility has outrun its 36-month estimate. Raw dispersion alone explains 4.4% of the next standardized factor range, but the forecast target is expressed in standardized units. When the outcome is instead the raw next-month range, raw dispersion explains 15.9% versus FDI's 7.0%, both computed on 716 identical dates. FDI is a complement to conventional measures, not a universally superior predictor.

FDI does not forecast the expected return of any single factor and does not replace a covariance model. A larger future range may reflect larger winner gains, larger loser losses, or both. Signed factor-selection returns provide a direct check of the limitation; their relation with FDI percentile is statistically indistinguishable from zero. The economically supported application is conditional risk awareness rather than an automatic mandate to increase allocation size.

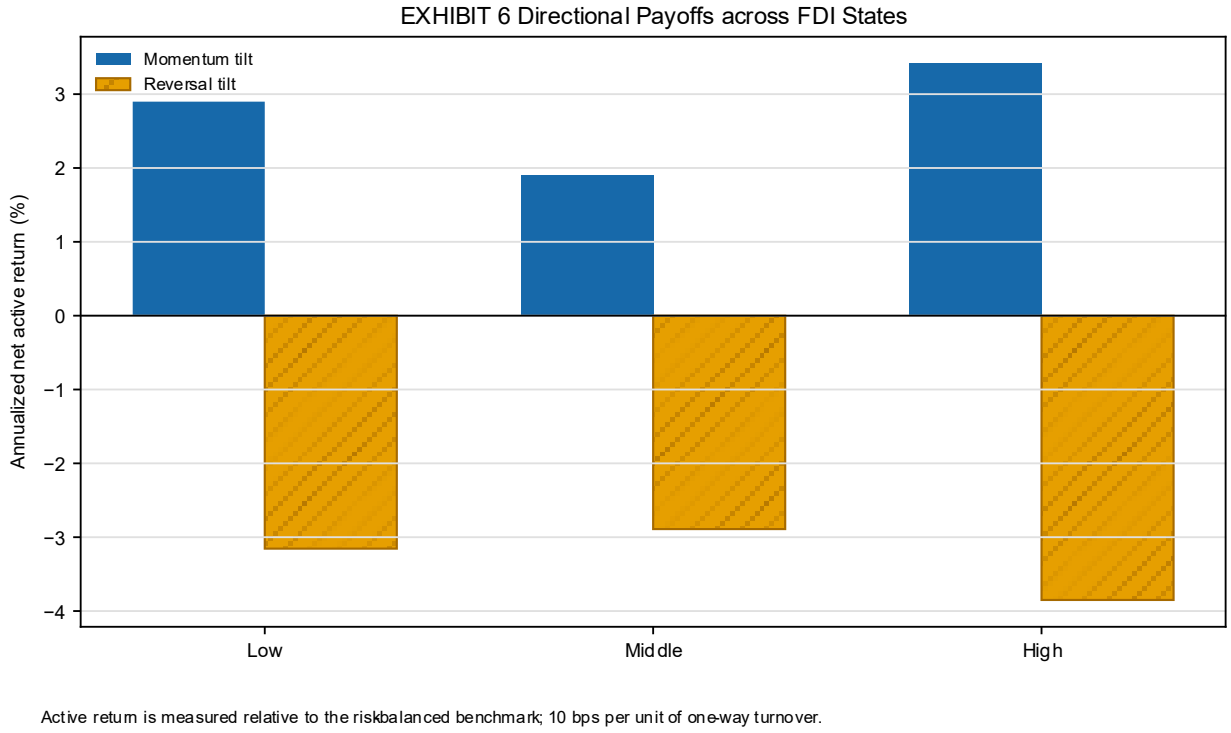
Momentum or Reversal after High Dispersion?

In this seven-factor sample, momentum rankings produce positive average payoffs, whereas reversal is the symmetric opposite-position counterfactual. Exhibit 4 reports annualized long–short selection spreads of 9.24%, 8.36%, and 4.24% in high, low, and middle FDI states. Only the first two are individually distinguishable from zero at conventional levels. Losses on the reversed diagnostic follow arithmetically from taking the negative of the same positions, rather than constituting independent evidence.

High dispersion does not establish a higher mean selection premium than low dispersion: the two state means are statistically indistinguishable. The signed momentum spread also has no discernible linear relation with the FDI percentile. Its absolute value, however, increases with that percentile. A larger FDI reading is therefore more informative about the potential size of a selection outcome than about choosing its profitable direction.

Exhibit 6 translates the diagnostic into long-only tilts around risk balance. Net of the standardized sleeve-turnover charge, the momentum tilt adds 2.90%, 1.91%, and 3.41% per year in low, middle, and high FDI states; reversal gives up 3.16%, 2.89%, and 3.85%, respectively. Although the high-state momentum point estimate is largest, it is not statistically distinguishable from the low-state mean. The monotonic feature is the increase in tracking error, not a demonstrated increase in directional alpha.

Exhibit 6. Directional Payoffs across FDI States



Notes: Active performance relative to the risk-balanced benchmark; returns are net of 10 bps per unit of one-way sleeve turnover.

Active risk is more systematically related to the FDI state than is average active return. Annualized tracking error of the unconditional momentum tilt relative to risk balance is 3.31%, 4.27%, and 7.13% in the low, middle, and high states. Net information ratios are 0.88, 0.45, and 0.48. Thus high FDI primarily signals greater dispersion in active outcomes; it does not provide an improvement in active return per unit of risk. The portfolio decision depends on signal confidence, liquidity, financing, mandate limits, and the institution's tracking-error tolerance.

Exhibit 7 provides a descriptive ranking decomposition, but it cannot establish that portfolio performance is independent of any one sleeve. The high standalone BAB Sharpe ratio and unequal frequency of BAB ranking outcomes warrant an explicit exclusion test. In the high-FDI state, the next-month return of a factor when it ranks among the top two exceeds its return when it ranks among the bottom two for all seven factors. Annualized continuation spreads range from 6.01% for RMW and 6.79% for SMB to 17.83% for BAB. MOM itself has a 14.03% continuation spread.

The low state also shows positive continuation for all seven sleeves, with particularly large effects for SMB and BAB. The middle state is weaker and mixed.

Exhibit 7. Factor-Level Continuation across FDI States

Next-month factor returns conditional on ranking among the top two or bottom two 12-month risk-adjusted momentum scores.

FDI State	Factor	Top-Rank Months	Top-Rank Next Return	Bottom-Rank Months	Bottom-Rank Next Return	Continuation Spread
Low	MKT	68	3.08%	60	0.31%	2.77%
Low	SMB	43	9.70%	89	-3.30%	13.01%
Low	HML	75	8.31%	60	-1.52%	9.82%
Low	RMW	51	2.24%	85	0.11%	2.13%
Low	CMA	71	3.16%	64	-0.27%	3.43%
Low	MOM	35	12.16%	88	10.96%	1.20%
Low	BAB	125	18.10%	22	-3.72%	21.82%
Middle	MKT	84	11.55%	52	12.33%	-0.78%
Middle	SMB	24	-2.16%	90	-0.70%	-1.46%
Middle	HML	36	10.15%	76	-5.08%	15.22%
Middle	RMW	66	5.47%	47	6.94%	-1.47%
Middle	CMA	51	4.13%	73	0.96%	3.17%
Middle	MOM	59	13.22%	50	7.81%	5.41%
Middle	BAB	102	7.74%	34	8.12%	-0.38%
High	MKT	61	13.12%	52	5.07%	8.05%
High	SMB	37	5.00%	94	-1.79%	6.79%
High	HML	37	12.22%	94	-1.17%	13.39%
High	RMW	69	5.15%	41	-0.86%	6.01%
High	CMA	36	12.83%	75	4.40%	8.44%
High	MOM	89	11.72%	28	-2.31%	14.03%
High	BAB	97	16.08%	42	-1.75%	17.83%

Notes: Returns are annualized averages of the next monthly factor return. These factor-level sorts are descriptive; ranking frequencies differ across factors and states.

This descriptive pattern provides limited support for a simple behavioral account dominated by immediate reversal. It is more consistent with gradual adjustment, changing factor risk premia, or persistent flows. Hong and Stein's (1999) underreaction framework provides one mechanism for momentum broadly, while the factor-momentum evidence of Ehsani and Linnainmaa (2022) suggests that common factor dynamics can transmit continuation across securities. Our seven-factor sample cannot distinguish among mechanisms. Portfolio construction does not require that distinction to be settled, but it does require the sign evidence to be respected.

From a State Variable to an Allocation Rule

The strategic anchor is an inverse-volatility factor portfolio. Diversifying return drivers within a portfolio can complement familiar asset-class diversification (Ilmanen and Kizer 2012), but the investor must first identify feasible factor sleeves. Each month, baseline sleeve weights are proportional to inverse lagged 36-month volatility. This transparent allocation is a practical approximation to risk parity when the aim is to prevent volatile sleeves from dominating. It avoids full-covariance weight optimization and follows the implementation logic discussed by Choueifaty and Coignard (2008), Maillard, Roncalli, and Teïletche (2010), and Clarke, de Silva, and Thorley (2013).

Momentum and reversal alter this anchor through centered ranks. For the momentum rule, the unscaled decision weight is

$$\widetilde{w}_{i,t}^{M} \propto b_{i,t}\left(1 + \kappa q_{i,t}\right),$$

where $b_{i,t}$ is the baseline inverse-volatility weight and $\kappa = 0.75$. The lowest-ranked factor retains 25% of its baseline weight, while the highest-ranked factor receives 175%; all weights remain nonnegative and are normalized to sum to one before portfolio-level volatility scaling. Reversal changes the plus sign to a minus sign.

The FDI-aware rule adds a fixed opportunity-strength multiplier a_t. It equals zero in the low state, 0.5 in the middle state, and one in the high state:

$$\widetilde{w}_{i,t}^{FDI-M} \propto b_{i,t}\left(1 + \kappa a_t q_{i,t}\right).$$

This schedule directly tests the intuitive policy proposed by many allocation teams: stay close to strategic balance when dispersion is low and permit larger active deviations when it is high. The thresholds, score window, and tilt strength are fixed in the reported design and not optimized against realized Sharpe ratios. The rule is a test of the common gating intuition, not a recommendation to increase active exposure after a high-FDI reading.

All portfolios target 10% annualized ex-ante total volatility from the lagged 36-month covariance matrix and cap leverage at 2.0. A uniform 10-basis-point charge per unit of one-way sleeve turnover facilitates comparisons; source factor returns are research series gross of their own trading and financing costs. Two additional strategies target a 4% ex-ante tracking-error budget around risk balance, subject to a cap of 1.25 times the ordinary momentum tilt. The constant-TE strategy divides the target by tracking error estimated from lagged covariance. The FDI-adjusted strategy multiplies that estimated risk by 0.80, 1.00, or 1.25 in low, middle, or high FDI before setting the tilt. In plain terms, high dispersion makes the same proposed active position look riskier,

so the FDI-adjusted rule scales it down; low dispersion reduces the forecast risk and permits a larger tilt, subject to the cap. Both strategies receive the same final total-volatility scaling. Realized TE need not equal 4%, and neither rule has been optimized for investment utility.

Exhibit 8 describes the rules and reports performance from July 1971 through March 2026, the period in which real-time FDI states are available. The net risk-balanced benchmark earns 8.72% per year with 7.45% realized volatility, a Sharpe ratio of 1.17, and a maximum drawdown of 29.24%. The unconditional momentum tilt earns 11.47% with 8.74% volatility and a Sharpe ratio of 1.31. Its maximum drawdown is 17.21%. Reversal earns only 5.43%, reduces the Sharpe ratio to 0.67, and suffers a 45.57% drawdown.

Exhibit 8. Allocation Rules and Portfolio Performance

Monthly implementation, July 1971-March 2026; 10% ex-ante volatility target, 2.0 leverage cap, and 10 bps per unit of one-way sleeve turnover.

Panel A: Pre-specified allocation rules

Strategy	Strategic Anchor	Active Rule	Rebalancing	Volatility Target	Leverage Cap
Risk-Balanced	Inverse lagged 36-month volatility	None	Monthly	10%	2.0
Momentum Tilt	Risk-balanced	Overweight high 12-month risk-adjusted ranks	Monthly	10%	2.0
Reversal Tilt	Risk-balanced	Overweight low 12-month risk-adjusted ranks	Monthly	10%	2.0
FDI-Aware Momentum	Risk-balanced	Momentum tilt strength 0/0.5/1 in low/middle/high FDI	Monthly	10%	2.0
FDI-Aware Reversal	Risk-balanced	Reversal tilt strength 0/0.5/1 in low/middle/high FDI	Monthly	10%	2.0
FDI-Informed TE Budget	Risk-balanced	4% TE target; FDI risk multipliers 0.80/1.00/1.25; cap 1.25x tilt	Monthly	10%	2.0
Constant TE Budget	Risk-balanced	4% TE target; constant risk multiplier 1.00; cap 1.25x tilt	Monthly	10%	2.0

Panel B: Net hypothetical performance

Strategy	Annual Return	Annual Volatility	Sharpe Ratio	Downside Deviation	Maximum Drawdown	Worst Month	Average One-Way Turnover	Start	End	Months
Risk-Balanced	8.72%	7.45%	1.17	4.28%	-29.24%	-10.94%	2.22%	1971-07-31	2026-03-31	657

Panel B: Net hypothetical performance

Strategy	Annual Return	Annual Volatility	Sharpe Ratio	Downside Deviation	Maximum Drawdown	Worst Month	Average One-Way Turnover	Start	End	Months
Momentum Tilt	11.47%	8.74%	1.31	4.97%	-17.21%	-10.49%	15.18%	1971-07-31	2026-03-31	657
Reversal Tilt	5.43%	8.09%	0.67	5.41%	-45.57%	-13.83%	15.58%	1971-07-31	2026-03-31	657
FDI-Aware Momentum	10.04%	8.23%	1.22	4.77%	-18.89%	-10.96%	18.51%	1971-07-31	2026-03-31	657
FDI-Aware Reversal	6.90%	8.07%	0.85	5.28%	-43.51%	-13.83%	18.81%	1971-07-31	2026-03-31	657
FDI-Informed TE Budget	12.06%	9.15%	1.32	5.23%	-17.57%	-11.14%	18.64%	1971-07-31	2026-03-31	657
Constant TE Budget	12.06%	9.26%	1.30	5.28%	-17.57%	-11.14%	18.54%	1971-07-31	2026-03-31	657

Panel C: Gross and net annualized return reconciliation

Strategy	Gross ann. return	Net ann. return	Annual cost (bp)	Monthly 1-way turnover
Risk-Balanced	8.75%	8.72%	2.7	2.22%
Momentum Tilt	11.65%	11.47%	18.2	15.18%
Reversal Tilt	5.61%	5.43%	18.7	15.58%
FDI-Aware Momentum	10.26%	10.04%	22.2	18.51%
FDI-Aware Reversal	7.12%	6.90%	22.6	18.81%
FDI-Informed TE Budget	12.28%	12.06%	22.4	18.64%
Constant TE Budget	12.28%	12.06%	22.3	18.54%

Panel D: Paired 12-month moving-block bootstrap confidence intervals for Sharpe differences

Difference	Point estimate	95% interval	Approx. p
Momentum - benchmark	0.142	[-0.049, 0.309]	0.151
FDI gate - benchmark	0.048	[-0.095, 0.193]	0.531
FDI gate - momentum	-0.094	[-0.168, -0.005]	0.036
TE budget - momentum	0.005	[-0.040, 0.057]	0.955
FDI TE vs constant TE	0.015	[0.002, 0.037]	0.011

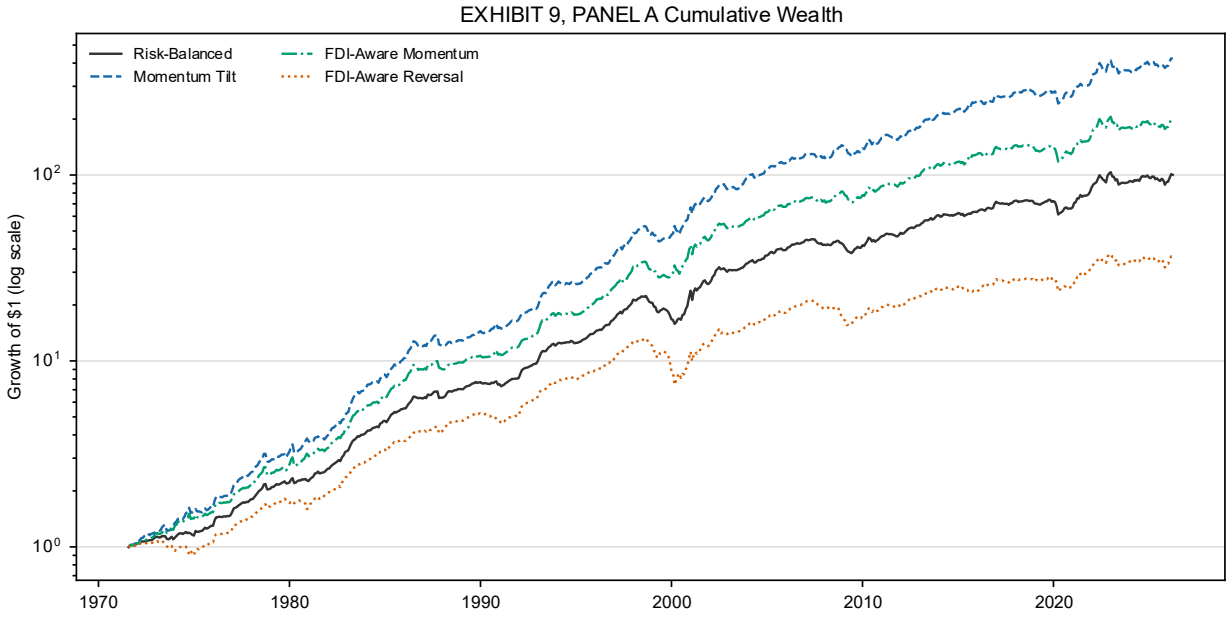
Notes: Panels A–B — The benchmark uses inverse lagged 36-month volatility, and active strategies change the corresponding long-only sleeve weights. Both TE strategies share the same 4% active-risk target and 1.25-times tilt cap; only the FDI-adjusted rule uses state multipliers (0.80/1.00/1.25) for estimated TE. All return series are hypothetical and exclude source-factor internal turnover, financing, management and capacity costs. Panel C — Annual cost equals the mean gross-minus-net return multiplied by 12; these are annual basis points of portfolio return, not turnover basis points. Both gross and net inputs derive from the same pre-cost factor research portfolios. Panel D — Paired 12-month moving blocks, 1,500 draws and fixed seed across comparable monthly series. Intervals and approximate two-sided p-values are descriptive in-sample measures of sampling uncertainty; the many strategy comparisons were not adjusted for multiple testing.

Exhibit 8 makes the attribution test explicit. The unconditional momentum tilt earns 11.47% net per year at 8.74% volatility (Sharpe 1.31). The constant-TE target earns 12.06% at 9.26% volatility (Sharpe 1.30), whereas its FDI-adjusted counterpart also earns 12.06%, with 9.15% volatility (Sharpe 1.32). Both suffer an approximately 17.57% maximum drawdown. Thus the roughly 0.59-percentage-point return improvement over unconditional momentum belongs almost entirely to TE targeting; adding FDI produces no material return increment. The FDI adjustment raises Sharpe by only 0.015 relative to constant-TE targeting. Its paired in-sample bootstrap interval is slightly above zero, but this small sample-dependent difference should not be construed as material economic outperformance. The FDI gate has a lower Sharpe (1.22) than unconditional momentum (1.31).

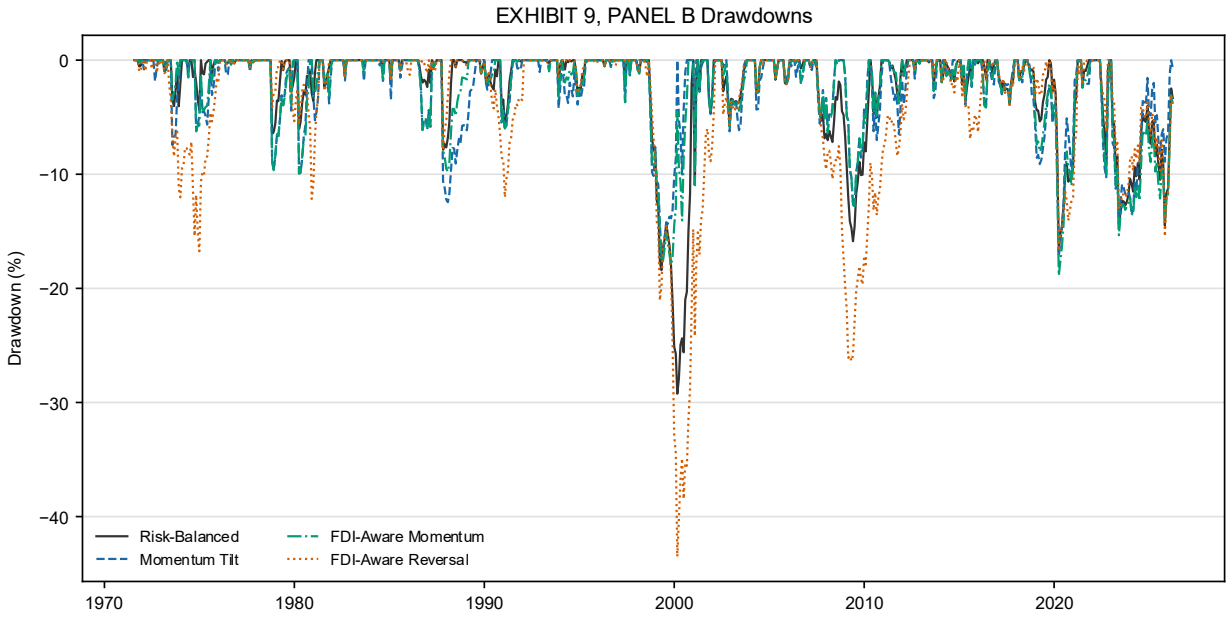
Panel C makes turnover attribution equally clear. Momentum gross and net annual returns are 11.65% and 11.47%, compared with 10.26% and 10.04% for the FDI gate. The gate's 18.51% monthly one-way turnover exceeds momentum's 15.18%, yet the resulting annual cost difference is only around four basis points—far too small to explain the 1.43-percentage-point net return gap. The gap chiefly reflects momentum exposure forgone in low-FDI states. Panel D also shows a statistically lower gate Sharpe than unconditional momentum (difference −0.094; 95% interval −0.168 to −0.005). By contrast, the confidence intervals for both momentum minus risk balance and the gate minus risk balance include zero. The higher full-sample maximum drawdown of risk balance (29.24%) comes largely from 1990–2006: after 2007 the gate's drawdown (18.89%) exceeds the benchmark's (16.84%).

Exhibit 9 plots net cumulative wealth and drawdowns. The wealth paths show the durable advantage of momentum direction and the large long-horizon cost of reversal. The drawdown panel shows that momentum-based allocations avoid part of the damage around the technology unwind and the global financial crisis, although neither rule eliminates factor losses. FDI-aware momentum sits between the strategic benchmark and unconditional momentum for most of the sample, which is exactly what its partial-activation design implies.

Exhibit 9. Cumulative Wealth and Drawdowns



Net hypothetical returns; 10% ex-ante volatility target, 2.0 leverage cap, and 10 bps per unit of one-way turnover.



Drawdowns are calculated from the net hypothetical wealth paths in Panel A .

Notes: Selected net hypothetical factor-allocation rules, July 1971-March 2026.

Economic Insights

The evidence separates three economic ideas that are often conflated. Opportunity width is not expected alpha. Directional predictability is not risk capacity. Risk capacity is not a forecast. FDI describes the first, factor momentum contributes to the second, and the investor must supply the third.

The key benefit of FDI is an additional gauge of outcome amplitude: in the matched active-portfolio comparison, it explains some absolute-return variation beyond trailing tracking error. FDI also retains a relationship with next-month standardized factor dispersion under short-volatility controls. These results accord with Shalen's (2025) emphasis on the sources of equity return dispersion, without assuming that stock-level insights automatically identify profitable factor timing.

Momentum direction is historically positive on average in the low- and high-dispersion states, but neither its advantage in portfolio Sharpe relative to risk balance nor its generality across sleeve menus is firmly established. Gradual adjustment, persistent institutional flows and changing factor premia are plausible explanations; Hong and Stein (1999) and Ehsani and Linnainmaa (2022) provide related mechanisms, while Bender et al. (2013) emphasize heterogeneous factor construction and implementation. The data do not identify a unique explanation.

A high-FDI environment produces a wider set of potential wins and losses and higher observed tracking error around active factor positions. It does not establish greater alpha per unit of active risk. That distinction matters to a manager facing short-horizon relative-performance limits: the warranted tilt could shrink even when the opportunity set widens. The near-identical returns of constant and FDI-adjusted TE strategies reinforce the need to judge state-dependent risk scaling against an equivalent strategy without the state adjustment.

Low FDI does not invalidate momentum. The gate's zero tilt in that state sacrifices the positive average returns earned by unconditional momentum and, over the full sample, reduces Sharpe. This is consistent with caution about conditional factor timing from Asness et al. (2017) and the estimation-risk concerns of DeMiguel, Garlappi, and Uppal (2009).

The result also clarifies the role of risk parity. A risk-balanced anchor is designed to distribute strategic risk, not to exploit every change in relative expected returns. FDI does not invalidate that anchor. It tells the manager when deviations from it will be more consequential. The proper comparison is therefore not “active versus passive” in the abstract. It is the incremental expected payoff, active volatility, drawdown tolerance, and implementation burden associated with leaving the anchor.

A Guide for Portfolio Managers

A factor-risk dashboard should start with the investor's own tradable sleeves, not standardized research portfolios. Factor definitions, neutralization, leverage, rebalance timing and embedded costs vary across suppliers. The construction used here is monthly, and different dealing frequencies require a separate history and volatility calibration.

The dashboard can show current FDI and its expanding percentile alongside raw cross-factor dispersion, prevailing VIX, trailing and model-implied active tracking error, liquidity and financing capacity. FDI is not a substitute for those measures: it conveys additional information about the width of possible standardized factor outcomes.

The return signal should appear separately: display each factor's momentum score, active weight and contribution to forecast tracking error. The committee can then distinguish a wide opportunity set from a dependable signal and detect unintended concentration in a sleeve such as BAB.

A high FDI percentile calls for reassessing the active-risk budget, not increasing it automatically. In the illustrative TE strategy, a high reading raises the projected tracking error of the same active weights by 25% and therefore scales down the tilt. The paired constant-TE comparison earns essentially the same return, with only a small Sharpe difference. Mandate constraints, confidence in the return signal, margin, funding and liquidity remain decisive.

A low FDI percentile should not turn off a separate momentum view; historical low-state momentum payoffs are positive. Hysteresis, no-trade bands and multiple consecutive state readings are possible operational conventions, but they are not tested here.

Ten basis points per unit of sleeve turnover is a uniform comparison charge, not a universal implementable hurdle. Financing, shorting, futures rolls, liquidity, swap fees, transaction costs within the source factors, and investor capacity may materially change net returns. The research BAB portfolio in particular may not match a fund’s obtainable exposures.

The practical sequence is to specify tradable factor sleeves and realistic costs; measure FDI and conventional volatility; select a separately tested return signal; estimate active tracking error; size exposures against the mandate; and review realized outcomes against risk balance, unconditional momentum and an otherwise identical budget without FDI. No one indicator determines the active budget.

Robustness and Limitations

Exhibit 10 separates parameter sensitivity, subsample stability and BAB concentration. Unconditional momentum is shown beside risk balance and the FDI gate, while the two matched TE-budget strategies are compared over the same three nonoverlapping calendar periods. The BAB-exclusion experiment rebuilds the six-factor FDI, expanding thresholds, ranks and weights from scratch. Relative Sharpe performance is less stable than unconditional return comparisons, and the historical factor menu materially affects conclusions.

Exhibit 10. Robustness and Subperiod Evidence

Panel A varies fixed signal parameters; Panel B reports all principal strategies over three nonoverlapping calendar subsamples. Panels C and D show the unconditional-momentum comparator and the re-estimated factor universe excluding BAB.

Panel A: Alternative specifications

index	Volatility Window	Momentum Window	Tilt Strength	FDI-Aware Momentum Return	FDI-Aware Momentum Sharpe	FDI-Aware Momentum Max Drawdown	High-FDI Momentum Active Return	FDI-Aware Reversal Return	FDI-Aware Reversal Sharpe	High-FDI Reversal Active Return	Benchmark Sharpe
0	24	6	0.75	9.63%	1.20	-21.05%	1.69%	6.71%	0.85	-3.12%	1.16
1	24	12	0.75	9.75%	1.20	-19.47%	2.23%	6.75%	0.86	-3.41%	1.16
2	36	6	0.75	9.87%	1.21	-21.08%	2.58%	6.92%	0.87	-3.65%	1.17
3	36	12	0.50	9.59%	1.24	-18.46%	2.19%	7.51%	0.96	-2.64%	1.17
4	36	12	0.75	10.04%	1.22	-18.89%	3.33%	6.90%	0.85	-3.94%	1.17
5	36	12	1.00	10.47%	1.17	-19.72%	4.42%	6.26%	0.75	-5.32%	1.17
6	60	12	0.75	9.98%	1.16	-19.88%	2.76%	6.79%	0.80	-3.68%	1.11
7	36	18	0.75	9.78%	1.20	-19.02%	2.78%	7.13%	0.89	-3.41%	1.17

Panel B: Net strategy performance by subperiod

Subperiod	Strategy	Annual Return	Annual Volatility	Sharpe Ratio	Maximum Drawdown	Months
1971-1989	Risk-Balanced	11.28%	6.24%	1.81	-7.86%	222
1971-1989	Momentum Tilt	14.92%	8.82%	1.69	-12.45%	222
1971-1989	FDI-Aware Momentum	13.17%	7.81%	1.69	-9.94%	222
1971-1989	FDI-Informed TE Budget	15.56%	9.53%	1.63	-13.85%	222
1971-1989	Constant TE Budget	15.51%	9.58%	1.62	-13.85%	222
1971-1989	FDI-Aware Reversal	9.20%	6.66%	1.38	-16.75%	222
1990-2006	Risk-Balanced	10.73%	8.47%	1.27	-29.23%	204
1990-2006	Momentum Tilt	13.31%	8.69%	1.53	-17.19%	204
1990-2006	FDI-Aware Momentum	11.89%	8.62%	1.38	-17.97%	204
1990-2006	FDI-Informed TE Budget	14.08%	8.80%	1.60	-17.45%	204

Panel B: Net strategy performance by subperiod

Subperiod	Strategy	Annual Return	Annual Volatility	Sharpe Ratio	Maximum Drawdown	Months
1990-2006	Constant TE Budget	14.14%	9.10%	1.55	-17.45%	204
1990-2006	FDI-Aware Reversal	8.62%	9.73%	0.89	-43.51%	204
2007-2026	Risk-Balanced	4.49%	7.42%	0.61	-16.83%	231
2007-2026	Momentum Tilt	6.52%	8.54%	0.76	-17.21%	231
2007-2026	FDI-Aware Momentum	5.39%	8.14%	0.66	-18.89%	231
2007-2026	FDI-Informed TE Budget	6.91%	8.93%	0.77	-17.57%	231
2007-2026	Constant TE Budget	6.91%	8.93%	0.77	-17.57%	231
2007-2026	FDI-Aware Reversal	3.17%	7.61%	0.42	-26.36%	231

Panel C: Unconditional momentum alongside principal parameter alternatives

Vol (m)	Rank (m)	Tilt	Mom net	Mom Sharpe	FDI gate net	Gate Sharpe
24	6	0.75	10.92%	1.27	9.63%	1.20
24	12	0.75	11.10%	1.27	9.75%	1.20
36	6	0.75	11.05%	1.29	9.87%	1.21
36	12	0.50	10.58%	1.32	9.59%	1.24
36	12	0.75	11.47%	1.31	10.04%	1.22
36	12	1.00	12.13%	1.27	10.47%	1.17
60	12	0.75	11.29%	1.24	9.98%	1.16
36	18	0.75	10.54%	1.23	9.78%	1.20

Panel D: Reconstructed six-factor universe excluding BAB

Strategy	Annual net return	Annual vol.	Sharpe	Max drawdown
Risk-Balanced	7.47%	7.00%	1.07	-19.86%
Momentum Tilt	9.55%	8.66%	1.10	-17.83%
FDI-Aware Momentum	8.41%	8.07%	1.04	-19.98%
FDI-Informed TE Budget	10.05%	9.05%	1.11	-17.93%

Notes: Panels A–B — Parameters are fixed within each specification, results are hypothetical and net of sleeve turnover charges unless identified as gross, and research factor returns do not include all underlying implementation frictions. Panel D — Dropping BAB changes the cross-sectional universe from seven to six factors; all FDI observations, expanding state thresholds, rankings, volatility scaling and realized portfolios are recomputed, rather than simply subtracting BAB's historical return from the seven-factor portfolios.

Exhibit 10 shows why full-sample performance is not a guarantee of downside protection. In 1971–1989, risk balance has Sharpe 1.81, exceeding both unconditional momentum and the gate

(about 1.69). From 1990–2006, unconditional momentum is the stronger directional comparator, and from 2007–2026 the gate suffers a deeper maximum drawdown (18.89%) than the benchmark (16.84%). Both TE-budget strategies have similar returns across each subperiod, reinforcing the limited role of the FDI multiplier in annual performance. Excluding BAB reduces momentum's Sharpe from 1.31 to 1.10 versus 1.07 for the six-factor risk-balanced benchmark; most of its portfolio-level advantage disappears. The FDI gate falls from 1.22 to 1.04 and does not overtake unconditional momentum.

The inputs are published research factor portfolios, not a unified set of live fund returns. BAB combines a high standalone historical Sharpe ratio with frequent top momentum rankings, and its omission weakens active-strategy performance considerably. Implementation requires a feasible proxy for each sleeve and investor-specific treatment of borrowing, shorting, roll costs, taxes, market impact, leverage and capacity. Our 10-basis-point sleeve charge is a common accounting hurdle, not a transaction-cost estimate for those underlying exposures.

Results depend on the factor menu. The 4% active-risk target and 0.80/1.00/1.25 state multipliers are fixed illustrations, not production thresholds or out-of-sample optimizers. The near-equality of constant and FDI-adjusted TE returns makes it especially important to evaluate a state-dependent scaling proposal against simpler budget rules before implementation.

FDI is a realized factor-level state rather than a structural shock, and its predictive relationship may reflect volatility clustering, correlation shifts, liquidity or investor flows. Results remain in-sample, and causal interpretation is not warranted. We use the Newey and West (1987) covariance estimator and paired block resampling for illustrative time-series uncertainty, but neither method resolves omitted predictors, search across specifications or the uncertainty of live execution.

The timing rules and sleeves are held fixed, but several reported measures remain sample-dependent. More elaborate nonlinear risk budgets, no-trade regions and signal blending might be useful and are not tested here. Related evidence on factor discovery and portfolio estimation error (Harvey, Liu, and Zhu 2016; DeMiguel, Garlappi, and Uppal 2009) counsels against presenting a modest in-sample Sharpe improvement as an investment recommendation.

Conclusion

The Factor Dispersion Index describes how widely returns to a specified set of factor sleeves differ after adjustment for prior volatility. It predicts next-month standardized factor-return dispersion and adds in-sample information about the absolute active return beyond trailing tracking error. A more widely separated factor menu should therefore prompt closer risk review, not an assumption of greater expected alpha.

A simple FDI gate loses profitable low-state momentum exposure and has a statistically lower Sharpe ratio than unconditional momentum in the sample. A constant 4% tracking-error budget and an otherwise identical FDI-adjusted budget both earn approximately 12.06% net annually; the FDI multiplier makes little difference to average returns. The portfolio Sharpe advantage of unconditional momentum over risk balance is itself statistically uncertain, and largely disappears when BAB is removed (1.10 versus 1.07). State-aware budgets require a stronger economic case than attractive isolated backtests.

For allocators, the defensible use of FDI is a portfolio-dashboard warning about the potential size of relative factor outcomes. It should sit beside raw dispersion, trailing tracking error, an independently evaluated directional model and a realistic assessment of risk tolerance, instruments and costs. Larger dispersion does not dictate larger positions; portfolio construction remains a separate decision.

Main Author Contributions

Dr. Xuan Feng, PhD, CFA, FRM, FDP, PMP, was the main author and primarily responsible for methodology design, data collection and quantitative analysis, manuscript writing, editing, and supervision. Dr. Feng currently works as a model risk quant in a leading global investment bank. And he holds a PhD in Finance from EDHEC Business School, France. His research interests include empirical asset pricing, portfolio construction, investment strategies, asset allocation, hedge funds, and financial risk management, with his work closely tied to practical investment problems — particularly return predictability, risk-controlled portfolio design, strategy evaluation, and the implementation of systematic investment approaches. A collection[i] of his research papers about quantitative finance, investment and portfolio strategies is available at https://github.com/XUAN-FENG9/Quant-Finance-and-Investment-Research. Dr. Feng is also a financial engineering practitioner with expertise in fixed income, quantitative pricing, and interest-rate derivatives. He has developed a widely used, publicly available GitHub tutorial repository (Rates-Pricing-Quant-Library) [ii] on interest-rate model pricing, serving as a comprehensive learning resource for interest-rate models and related derivative pricing frameworks — reflecting his broader interest in bridging rigorous quantitative finance theory with practical model implementation. Further information is available on the author's LinkedIn profile[iii].

Author Statement

The research was conducted independently outside the authors' employment and previous universities/schools. The views expressed are solely those of the authors and do not represent any employer, company, university, or other institution.

The authors report no conflicts of interest, and this research received no financial support from any public, commercial, academic, or not-for-profit organization.

OpenAI ChatGPT was used for enhancing analysis codes, language editing, and structural refinement. The authors independently reviewed and verified the data, methods, source code, numerical results, references, interpretations, and conclusions and accepted full responsibility for the manuscript.

Data Availability

The source series and documentation are available from the Kenneth R. French Data Library (https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html), AQR Data Library (https://www.aqr.com/Insights/Datasets) and FRED (https://fred.stlouisfed.org/series/VIXCLS).

Replication Package

All data and data-processing scripts, analysis code, robustness tests, and replication files are available in the following GitHub repository: [JBIS: Factor Dispersion][iv]. The empirical results reported in this paper are fully replicable using the data and code provided in the repository. The notebook (.ipynb) reproduces the quantitative results after the user changes a single local DATA_DIR setting.

Important Investment Disclaimer

This article is provided for research and informational purposes only. It does not constitute investment advice, an offer, or a recommendation to purchase or sell any security. Historical results and the ex-post allocation illustrations are not forecasts.

References

Arnott, Rob, Campbell R. Harvey, Vitali Kalesnik, and Juhani T. Linnainmaa. 2019. "Alice's Adventures in Factorland: Three Blunders That Plague Factor Investing." *The Journal of Portfolio Management* 45 (4): 18–36.

Asness, Clifford S., Swati Chandra, Antti Ilmanen, and Ronen Israel. 2017. "Contrarian Factor Timing Is Deceptively Difficult." *The Journal of Portfolio Management* 43 (5): 72–87.

Asness, Clifford S., Tobias J. Moskowitz, and Lasse Heje Pedersen. 2013. "Value and Momentum Everywhere." *The Journal of Finance* 68 (3): 929–985.

Barroso, Pedro, and Pedro Santa-Clara. 2015. "Momentum Has Its Moments." *Journal of Financial Economics* 116 (1): 111–120.

Bender, Jennifer, Remy Briand, Dimitris Melas, and Raman A. Subramanian. 2013. "Foundations of Factor Investing." *MSCI Research Insight*, December.

Carhart, Mark M. 1997. "On Persistence in Mutual Fund Performance." *The Journal of Finance* 52 (1): 57–82.

Cederburg, Scott, Michael S. O'Doherty, Feifei Wang, and Xuemin Yan. 2020. "On the Performance of Volatility-Managed Portfolios." *Journal of Financial Economics* 138 (1): 95–117.

Choueifaty, Yves, and Yves Coignard. 2008. "Toward Maximum Diversification." *The Journal of Portfolio Management* 35 (1): 40–51.

Clarke, Roger, Harindra de Silva, and Steven Thorley. 2013. "Risk Parity, Maximum Diversification, and Minimum Variance: An Analytic Perspective." *The Journal of Portfolio Management* 39 (3): 39–53.

Daniel, Kent, and Tobias J. Moskowitz. 2016. "Momentum Crashes." *Journal of Financial Economics* 122 (2): 221–247.

DeMiguel, Victor, Lorenzo Garlappi, and Raman Uppal. 2009. "Optimal versus Naive Diversification: How Inefficient Is the 1/N Portfolio Strategy?" *The Review of Financial Studies* 22 (5): 1915–1953.

Ehsani, Sina, and Juhani T. Linnainmaa. 2022. "Factor Momentum and the Momentum Factor." *The Journal of Finance* 77 (3): 1877–1919.

Fama, Eugene F., and Kenneth R. French. 1993. "Common Risk Factors in the Returns on Stocks and Bonds." *Journal of Financial Economics* 33 (1): 3–56.

Fama, Eugene F., and Kenneth R. French. 2015. "A Five-Factor Asset Pricing Model." *Journal of Financial Economics* 116 (1): 1–22.

Frazzini, Andrea, and Lasse Heje Pedersen. 2014. "Betting against Beta." *Journal of Financial Economics* 111 (1): 1–25.

Gupta, Tarun, and Bryan T. Kelly. 2019. "Factor Momentum Everywhere." *The Journal of Portfolio Management* 45 (3): 13–36.

Harvey, Campbell R., Yan Liu, and Heqing Zhu. 2016. "… and the Cross-Section of Expected Returns." *The Review of Financial Studies* 29 (1): 5–68.

Hong, Harrison, and Jeremy C. Stein. 1999. "A Unified Theory of Underreaction, Momentum Trading, and Overreaction in Asset Markets." *The Journal of Finance* 54 (6): 2143–2184.

Ilmanen, Antti, and Jared Kizer. 2012. "The Death of Diversification Has Been Greatly Exaggerated." *The Journal of Portfolio Management* 38 (3): 15–27.

Jegadeesh, Narasimhan, and Sheridan Titman. 1993. "Returns to Buying Winners and Selling Losers: Implications for Stock Market Efficiency." *The Journal of Finance* 48 (1): 65–91.

Maillard, Sébastien, Thierry Roncalli, and Jérôme Teïletche. 2010. "The Properties of Equally Weighted Risk Contribution Portfolios." *The Journal of Portfolio Management* 36 (4): 60–70.

McLean, R. David, and Jeffrey Pontiff. 2016. "Does Academic Research Destroy Stock Return Predictability?" *The Journal of Finance* 71 (1): 5–32.

Moreira, Alan, and Tyler Muir. 2017. "Volatility-Managed Portfolios." *The Journal of Finance* 72 (4): 1611–1644.

Moskowitz, Tobias J., Yao Hua Ooi, and Lasse Heje Pedersen. 2012. "Time Series Momentum." *Journal of Financial Economics* 104 (2): 228–250.

Newey, Whitney K., and Kenneth D. West. 1987. "A Simple, Positive Semi-Definite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix." *Econometrica* 55 (3): 703–708.

Shalen, Catherine. 2025. "The Roots of Dispersion." The Journal of Portfolio Management 51 (9): 157–165. https://doi.org/10.3905/jpm.2025.1.734.

Solnik, Bruno H., and Jacques Roulet. 2000. "Dispersion as Cross-Sectional Correlation." *Financial Analysts Journal* 56 (1): 54–61.

Stivers, Chris, and Licheng Sun. 2010. "Cross-Sectional Return Dispersion and Time Variation in Value and Momentum Premiums." *Journal of Financial and Quantitative Analysis* 45 (4): 987–1014.

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i https://github.com/XUAN-FENG9/Quant-Finance-and-Investment-Research

ii https://github.com/XUAN-FENG9/Rates-Pricing-Quant-Library

iii https://www.linkedin.com/in/dr-xuan-feng-217556a4/

iv https://github.com/XUAN-FENG9/Quant-Finance-and-Investment-Research/tree/main/JBIS_Factor_Dispersion_and_Active_Risk_Budgeting